

Relationship Between Lowest Vaccination Rate Among Six Basic Vaccines and Child Mortality in 2020

Report based on 2020 country-level data

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This report examines the relationship between two global health indicators in 2020:

- Lowest vaccination rate among six basic vaccines
- Child mortality, measured as deaths per 1000 live births

The report includes a brief data inspection, 2020 subsetting, map-input checks, two world maps, a scatterplot, interpretation, and a reflection section.

1. Introduction

Vaccination coverage is an important public-health indicator because vaccines help prevent serious childhood illnesses. Child mortality is also a key indicator of population health and health-system performance. This report compares these two indicators at country level in 2020 to explore whether countries with lower vaccination coverage also tend to have higher child mortality.

2. Brief Data Inspection

Before analysis, both datasets were briefly viewed to understand their structure, variable availability, and dimensions.

Dataset	Dimensions
Vaccination dataset	129 rows x 30 columns
Child mortality dataset	194 rows x 303 columns

3. Subsetting to 2020 and Map Input Checks

Both datasets were subset to 2020. Before creating the maps, the 2020 data were checked to ensure that each country appeared only once.

- Vaccination 2020 data available for: 122 countries
- Child mortality 2020 data available for: 194 countries
- Duplicate country rows in vaccination 2020 data: 0
- Duplicate country rows in child mortality 2020 data: 0
- Countries included in the final scatterplot: 122

The 2020 map inputs were in the correct format, with one row per country.

4. World Maps for 2020

The two maps below show the geographic distribution of vaccination coverage and child mortality in 2020.

Lowest vaccination rate among six basic vaccines, 2020
(percentage, by country)

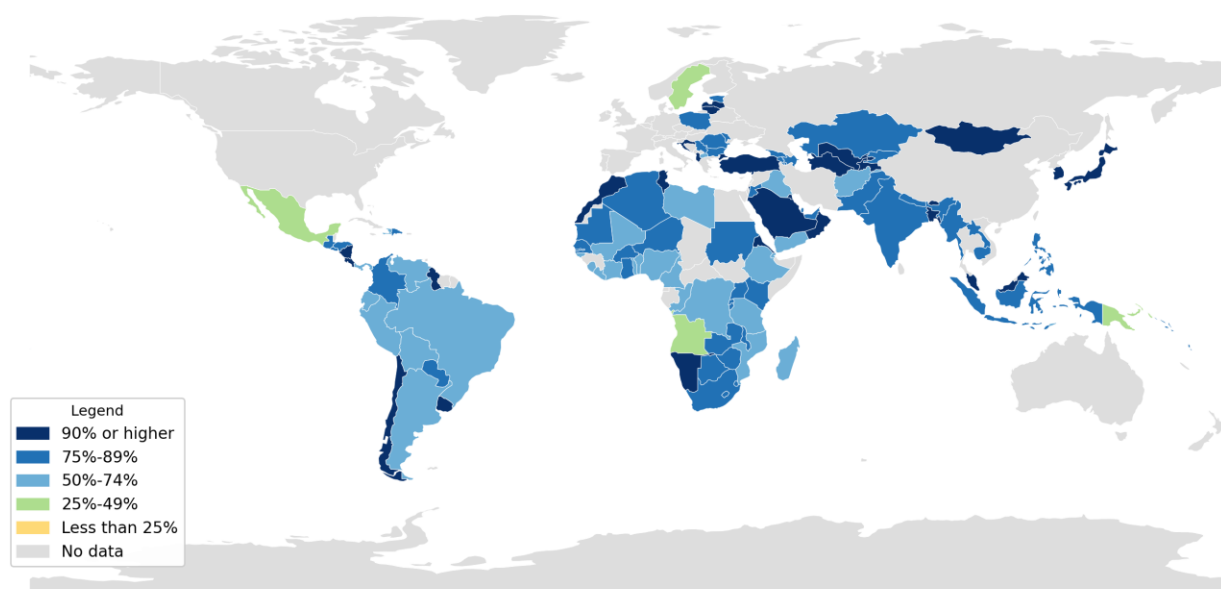


Figure 1. Lowest vaccination rate among six basic vaccines, 2020.

Higher vaccination coverage is concentrated in many higher-income regions, while lower minimum coverage appears in several parts of sub-Saharan Africa and other vulnerable settings.

Child mortality, 2020
(deaths per 1000 live births, by country)

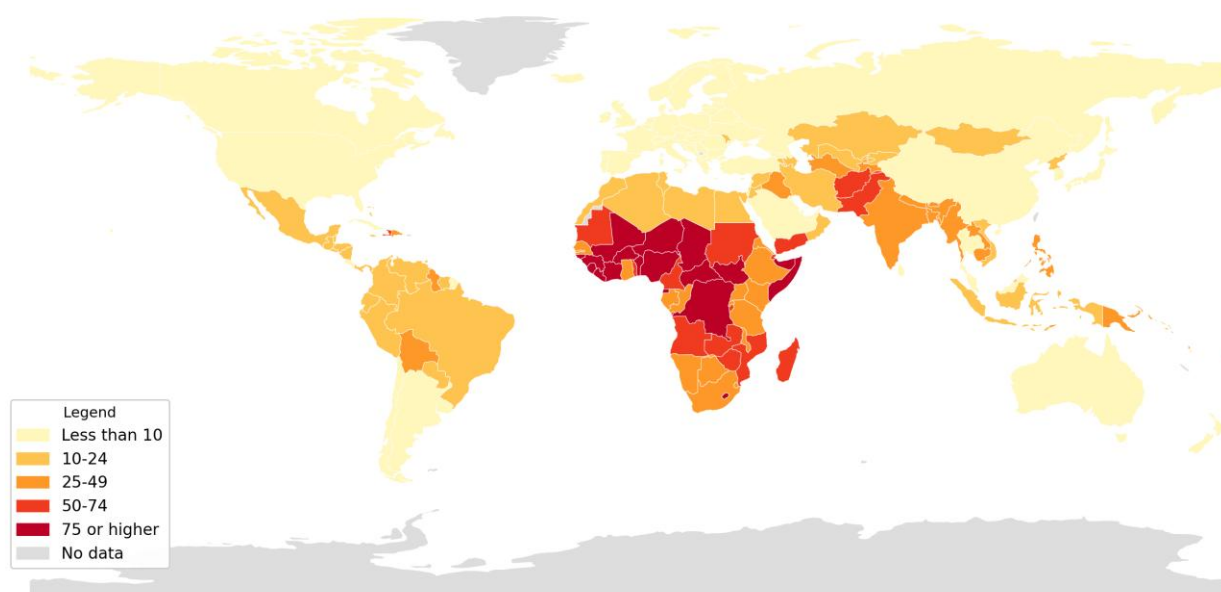


Figure 2. Child mortality, 2020 (deaths per 1000 live births).

Higher child mortality is concentrated in several low-income and conflict-affected regions, especially parts of sub-Saharan Africa. Lower mortality is more common in Europe, North America, and other higher-income regions.

Together, the maps suggest that countries with weaker vaccination coverage often also face higher child mortality.

5. Scatterplot and Relationship

Each point represents a country with data for both indicators in 2020.

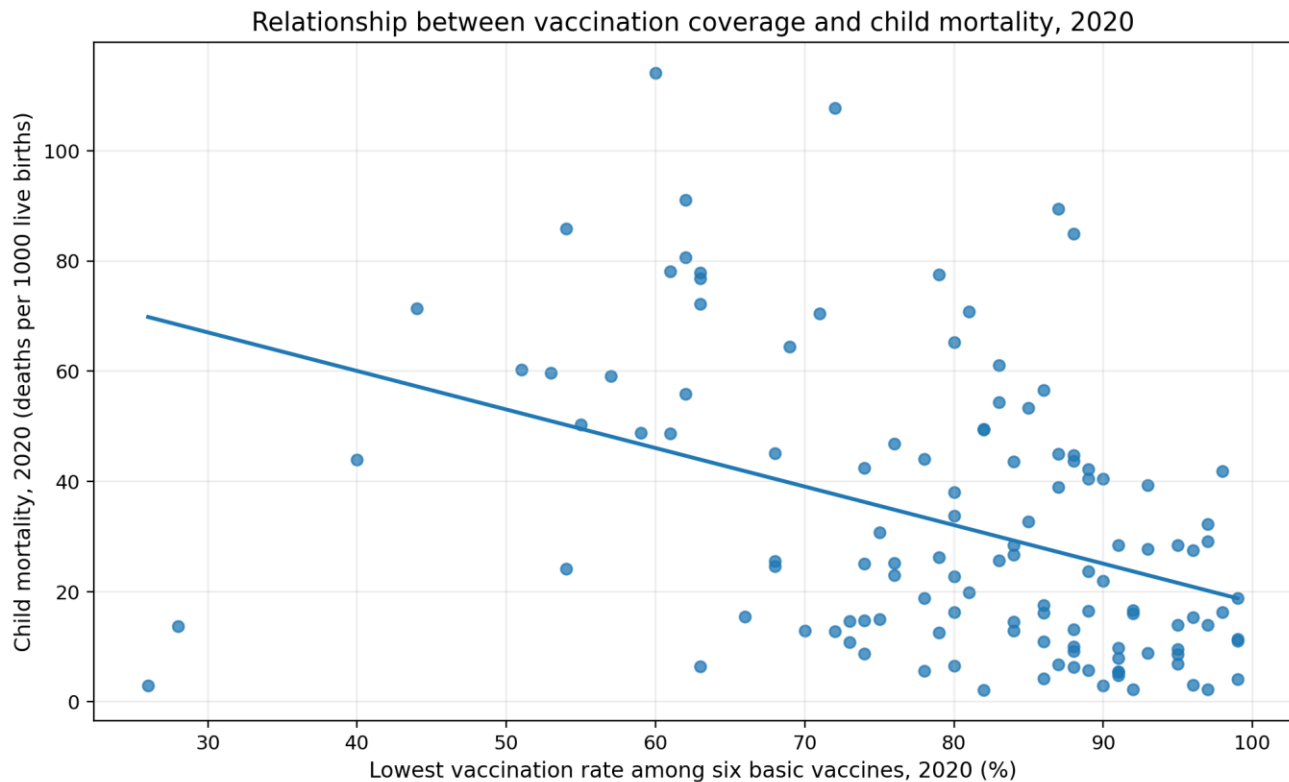


Figure 3. Relationship between lowest vaccination rate and child mortality, 2020.

6. Brief Interpretation

- Pearson correlation: approximately -0.40
- The relationship is moderately negative.
- Countries with lower vaccination coverage generally tend to have higher child mortality.
- The fitted line slopes downward, suggesting that higher minimum vaccination coverage is generally associated with lower child mortality.
- This is an association, not proof of causation, because mortality is also influenced by healthcare access, nutrition, sanitation, income, conflict, and other public-health conditions.

7. Reflection

Working on this assignment to analyze data using LLMs has been incredibly enlightening. I used ChatGPT to analyze the data and generate the report, which took the model 1 minute and 21 seconds to complete and just 5 seconds to refine. This efficiency is quite impressive.

What I found particularly interesting was observing the background thought process the model employed to generate the report. I noted its speed, depth of analysis, error-checking, and re-attempts. This experience further confirmed the importance of collaborating with AI, as surpassing its capabilities is nearly impossible for humans.

Overall, I am very impressed and look forward to leveraging the opportunities that AI presents.

End of report.